

MADHU RAMAKRISHNAPPA

SF Bay Area madhuramakrishnappa4@gmail.com +1 (617) 238-8940 in/mrk16

SUMMARY

Full Stack Software Engineer with **4+ years of experience** delivering secure, scalable applications in the **financial services domain**. Skilled in **Java (Spring Boot)** and **MERN stack** for building microservices and dynamic front ends across banking and investment platforms. Proficient in **AWS, CI/CD**, and cloud security, with a strong focus on compliance, performance, and system reliability.

SKILLS

Programming Languages & Libraries: Java, JavaScript, Python (pandas, NumPy), SQL, HTML, CSS
Frameworks & Tools: Spring Boot, Node.js (Express), React, Angular, Maven, Postman
Databases: MongoDB, MySQL, PostgreSQL, SQL Server, Snowflake
Cloud & DevOps: AWS (EC2, S3, Lambda, VPC, IAM, CloudTrail), Docker, Jenkins, GitHub Actions
Security & Compliance: JWT Authentication, IAM Roles, VPC Isolation, SOC 2 alignment
Testing & CI/CD: JUnit, Postman, Git, CI/CD Pipelines (Jenkins, GitHub Actions)
Architecture & Design: Microservices, REST APIs, MVC Architecture, Object-Oriented Programming, Modular Design
Collaboration Tools: Jira, MS Project, Zoho CRM, Google Colab, Jupyter Notebook, PyCharm, VS Code

EXPERIENCE

Software Development Engineer

Northern Trust June 2024 - Present, USA

- Developed and maintained full-stack financial applications using Java (Spring Boot) and MERN stack, contributing to the successful processing of over 2.5 million secure transactions monthly for private wealth clients.
- Engineered and optimized RESTful APIs to support real-time banking workflows, reducing backend response latency by 28% and enabling sub-150ms API performance for critical services.
- Designed a hybrid data storage model with MongoDB and MySQL, improving data retrieval performance by 20% and reducing API payload sizes by 18% in high-traffic financial modules.
- Led deployment of full-stack applications to AWS EC2 and Lambda, cutting infrastructure provisioning time by 42% and improving system uptime to 99.95% across production environments.
- Implemented cloud security best practices using IAM roles, JWT authentication, and VPC isolation, achieving 100% adherence to SOC 2 Type II controls in quarterly compliance audits.

Full Stack Java Developer (Client: Equinity)

Tata Consultancy Services January 2021 - August 2023

- Developed robust backend services using Java (Spring Boot) and MySQL to support Equinity's corporate equity management system, handling over **500K daily transactions** with guaranteed data accuracy.
- Re-architected legacy finance modules into modular **microservices**, reducing system complexity by **21%** and enabling faster feature rollout across equity and dividend platforms.
- Migrated legacy applications to **AWS**, leveraging **EC2, S3**, and **IAM** to reduce hosting costs by **23%** and improve infrastructure scalability and uptime.
- Created secure **REST APIs** for managing stock plans, vesting schedules, and employee grant data, cutting manual processing time by **55%** and increasing system consistency.
- Automated backend build and deployment using **Jenkins, Docker**, and **Maven**, reducing release errors and accelerating delivery timelines for business-critical applications.
- Collaborated with **QA** and business teams to integrate testing suites and perform regression testing using **Postman**, ensuring smooth deployment of finance features across multiple environments.

EDUCATION

Master of Science, Information Technology
University of Massachusetts Boston • USA • 2025

Bachelor of Engineering, computer science Engineering
Visvesvaraya Technological University • India • 2022

PROJECT

ESG Portfolio Analyzer – Investment Risk & Sustainability Dashboard

University of Massachusetts Boston • February 2025 - May 2025

- Designed a system that helps investors identify high-risk or low-performing ESG assets by correlating sustainability scores with real-time market indicators using a Java + MERN-based tech stack.
- Structured a hybrid data model using MongoDB for ESG datasets and MySQL for financial performance metrics, supporting side-by-side comparisons of over 1,000 equities.
- Automated ingestion of ESG and market data from third-party APIs using AWS Lambda, enabling live dashboard updates every 12 hours for real-time visibility.

Credit Risk Analysis Using Machine Learning

Visvesvaraya Technological University • December 2021 - March 2022

- Engineered ML models using scikit-learn to predict loan defaults with 87% AUC-ROC, improving risk identification by 25% over logistic baseline.
- Preprocessed and analyzed 10,000+ loan records using pandas, optimizing features such as credit history length, debt-to-income ratio, and employment stability.
- Built Tableau dashboards that visualized risk segmentation across five borrower categories, enhancing interpretability for non-technical stakeholders.

CERTIFICATIONS

AWS - Cloud Security Foundation
University of Massachusetts Boston • 2025